

Home Assignment 2

Question 1

1. Use the haifa_res.xls file. Fit a linear model of the monthly spending as a function of monthly income differentiated by sex. Consider both an indicator variable for sex and an interaction term.
 - a. Present the theoretical model and list the assumptions. Present fitted model and its summary. Is the model significant?
 - b. Plot the results (one line for males and another for females).
2. Perform a statistical test to check whether the relationship between income and spending differs by sex. Formulate the tested hypothesis. Present the test statistic. Discuss the results.
3. Use the global best approach with the adjusted R-squared criteria to choose the best model. Summarise your results.
4. Use the stepwise, backwards, and forward algorithms with the AIC criteria to choose the best model. Summarise the results. For each method, present the initial and the final models. Does the best model differ as a function of the selection method?
5. Use the relaxed LASSO algorithm (with a 10-fold CV for the tuning parameter selection) to find the best model. Present the final model. Report and compare your results to the AIC criterion stepwise regression best model.

Question 2

Read the psych_data.txt file. It contains data on 999 Haifa, Tel-Aviv, and Jerusalem students who took the psychometric exam. The variables are:

psych - the exam grade (integer value between 200 and 800)

city - the city where the student took the exam (Haifa, Tel-Aviv, and Jerusalem)

income – the reported monthly income of the student's family (in thousands of ILS)

age - student's age when they took the exam

Our task is to find the best possible model that predicts the psychometric exam grades using family income, city, and age.

1. Read the file and provide a concise summary of the data. Both visual and numeric.
2. Use only the available variables, without interactions, to find the best model using the all-subset (global best) approach with the adjusted R-squared criteria. Summarise the results. Present explicitly the best mode.
3. Add all pairwise interactions between city and income, and consider nonlinear functions of income. Which nonlinear transformations are more plausible? Concave or Convex? According to your answer, add these transformations as new variables to your dataset.
4. Choose the best model using (1) Stepwise AIC and (2) leave-one-out CV (MSE). Compare the results. Perform model diagnostics of the final model.
5. Use the relaxed LASSO algorithm (with a 10-fold CV for the tuning parameter selection) to find the best model. Report and compare your results to the AIC criterion stepwise regression best model.

Question 3

1. For the titanic data set. For the full model that contains the class, the sex, age and the interaction term between sex and age - use the nonparametric bootstrap to compute the 95%-level CIs of all the model coefficients. Plot the bootstrapped distributions and compare the results to the analytical CIs from the standard model output.
2.
 - a. Use the Elastic net algorithm with $\alpha = 0.7$, and a 15-fold CV for tuning parameter selection. Report and compare your results to the AIC criterion stepwise regression best model.
 - b. Use the nonparametric BS to construct 95%-level CIs for the non-zero coefficients of the final model. Summarise the results.

Question 4

Let $y_i = \beta x_i + \epsilon_i$, $i = 1, \dots, n$, where $\mathbb{E}[\epsilon_i|x] = 0$, $\text{Var}(\epsilon_i|x) = \sigma^2$, and $\text{cov}(\epsilon_i, \epsilon_j) = 0$ for all $i \neq j$.

1. Find the OLS estimator, its expected value, variance and show its consistency.
2. Find the Ridge estimator where $\beta^2 \leq r$, its expected value, variance and show its consistency.
3. Find the LASSO estimator where $|\beta| \leq r$, its expected value, and show its consistency.

Question 5

The AIC is defined as

$$AIC = -2 \log(\text{Likelihood}) + 2p$$

where p is the number of model parameters. Show that for the linear multiple regression model

$$y_i = \beta_0 + \beta_1 x_{1i} + \dots + \beta_p x_{pi} + \epsilon_i, i = 1, \dots, n$$

with the classical assumptions on the error term, i.e.,

$$\epsilon_i \sim^{iid} N(0, \sigma^2), i = 1, \dots, n$$

boils down to

$$AIC = n \log(\hat{\sigma}_{MLE}^2) + C_{n,p}$$

where $\hat{\sigma}_{MLE}^2$ is the maximum likelihood estimator of σ^2 , and $C_{n,p}$ is a term that depends only on the sample size n and the number of parameters p .

Question 6

- a. Let $X \sim DU[1, N]$. Derive the expected value and the variance of X .
- b. Let n_1 and n_2 be the number of patients in the first and the second group, respectively. Let Y be the outcome variable, and R_1, R_2 be the rank sum in each group, respectively.
 - b.1 Write $R_1 + R_2$ in terms of N , where $n_1 + n_2 = N$.
 - b.2 Find $\mathbb{E}[R_1]$ and $\mathbb{E}[R_2]$ under the null hypothesis of no differences between the groups medians.